

Practical 8. Functions and Algorithms

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In this practical, you will write one Python functions and one class. You may also need to look for online references to complete this assignment, which is good practice on how to ask right questions. Please don't be shy about asking questions.

You are expected to use your GitHub account from previous weeks to store your code. Please create a directory called Practical 8 in your GitHub repository and put all your python scripts from this practical there.

1 Learning objectives

- Write simple functions in Python
- Define and use classes in Python
- Read and understand a simple piece of code

2 Predicting protein mass

Understanding the mass of a protein is important in drug discovery, as it helps us to better understand the protein's interactions with potential therapeutic agents.

Write a function which takes an amino acid sequence and returns the mass of the total protein in atomic mass units (amu). Remember the word 'return' has a specific meaning when referring to functions in Python. Your function should check that all supplied amino acids are defined correctly and can be found in this table. Make sure that your code reports an error if a supplied amino acid has no recorded mass.

The weight of each residue is shown in the table below:

Amino acid	Symbol	Residue Mass, amu (monoisotopic)
Glycine	G	57.02
Alanine	A	71.04
Serine	S	87.03
Proline	P	97.05
Valine	V	99.07
Threonine	T	101.05
Cysteine	C	103.01
Isoleucine	I	113.08
Leucine	L	113.08
Asparagine	N	114.04
Aspartic Acid	D	115.03
Glutamine	Q	128.06
Lysine	K	128.09
Glutamic Acid	E	129.04
Methionine	M	131.04
Histidine	H	137.06
Phenylalanine	F	147.07
Arginine	R	156.10
Tyrosine	Y	163.06
Tryptophan	W	186.08

Please include an example of how this function should be called within your code.

3 Nutrition Data Tracker

We are going to write some code to keep track of an individual's nutritional input.

To do this, you will first create a new Python class called 'food_item' that contains the name of the food item and counts of its calories, protein, carbohydrates and fat. For example, an apple might have 60 calories, 0.3 g protein, 15 g carbohydrate, 0.5 g fat.

Next you will write a function that takes a list of items that have been defined in your class that an individual has consumed over a 24 hr period. Your function should calculate and report the total calories, protein, carbohydrate and fat consumed over that period. It should also report a warning if the individual has consumed more than 2,500 calories or 90 g fat.

Please include an example of using this class and function within your code.

4 For your portfolio

Protein mass predictor

- The marker will check that a function has been defined properly as described in the lecture.
- The marker will check that there is an example function call contained within this script.
- The marker will input a new amino acid sequence to your function to determine whether the correct protein mass or error message is returned.

Nutrition Data Tracker

- The marker will supply a list of food that have been consumed over 24 hrs to determine whether your code correctly updates and, when necessary, reports a warning of excessive calorific or fat consumption.

You can add or edit things after the Practical and Tutorials have finished. We do not look at the commit date, we just want it all to be there!